

SAI SUDHEER VISHNUMOLAKALA

☎ (813)203-3465 📄 Exp: 4 Years 📍 Tampa, FL (Open to relocate)

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EXPERIENCE

Data Scientist - Graduate Research Assistant

Feb 2024 – Present

University of South Florida

Tampa, Florida

- **Automated Data Pipelines:** Migrated Excel-based workflows to Python-driven **ETL pipelines** with PostgreSQL, reducing processing time by **80%**. Built **ML models** to analyze defense agreements, extracting insights for strategic decision-making.
- **Data Visualization:** Developed a web-based interactive dashboard using Dash, Flask, and Plotly, enabling real-time data visualization and improving accessibility for stakeholders. Increased user engagement with data insights by 40%.
- **ML:** Engineered a token-level anomaly detection system utilizing an **Autoencoder model**, assigning probability scores to detect anomalous command patterns. Achieved an **AUC-ROC of 0.94**, strengthening cybersecurity defenses against injection attacks.
- **Gen AI:** Innovated a **chatbot** for USF, integrating web scraping, **vector databases**, and OpenAI embeddings with **LangChain**. Leveraged **RAG models** to generate accurate, context-aware responses.

Data Scientist

May 2022 – Jun 2023

Tiger Analytics

- Developed a real-time fraud detection model using **XGBoost**, reducing fraudulent transactions by 35% and preventing financial losses of \$5.2 million annually.
- Built unsupervised learning models (**K-Means**, **DBSCAN**) to segment customers, enabling personalized marketing strategies that increased loan conversion rates by 20%.
- **Cloud:** Developed and deployed ML models on **AWS SageMaker**, leveraging Redshift for data storage and Glue for ETL. Automated pipelines and model training, cutting processing time by 30% and enabling real-time predictions.
- **Advanced SQL:** Expertly utilized advanced SQL techniques, including CTEs, window functions, views, and stored procedures to optimize query performance, enhancing the efficiency of ETL pipelines and supporting complex data operations at scale.

Data Analyst

Jul 2021 – May 2022

Tata Consultancy Services

- Created Power BI dashboards with over 50+ **KPIs** for warehousing and customer service, enabling real-time tracking of operational metrics and facilitating data-driven decision-making.
- Implemented automated **feature engineering** for predictive models, improving model accuracy by 18% while reducing **training time by 35%**, leading to faster and more reliable predictions.

NLP Researcher (Full-time Intern)

Feb 2020 – Dec 2020

LTRC Lab, IIIT Hyderabad

- Led a research project on Abstractive Summarization for Telugu (Indian language) datasets, utilizing a deep reinforced model combining supervised learning and reinforcement learning, achieving a ROUGE score of 0.34.

EDUCATION

University of South Florida

Tampa, FL

Master of Science in **Artificial Intelligence and Business Analytics**; GPA: 4.0

Aug 2023 – May 2025

- **Coursework:** Data Warehousing, Big Data, Text Analytics, Data Science, Distributed Information Systems, Advanced System Analysis and Design, Analytical Methods for Business, Data Mining, Project Management, Enterprise Information Systems

TECHNICAL SKILLS

Languages & Frameworks: Python, R, Java, SQL, TensorFlow, PyTorch

Machine Learning: Supervised & Unsupervised Learning, Ensemble Models, Neural Networks, Text Analytics

Gen AI: Prompt Engineering, Parameter-efficient Fine Tuning, RAG, Langchain, LLM

Cloud & Big Data: AWS, Databricks, Docker, Apache Spark, NoSQL

Data Engineering: ETL Pipelines, Data Warehousing, Data Modeling, Snowflake, Cassandra, PostgreSQL

Statistics: Probability, Hypothesis Testing, Regression Analysis, Statistical Modeling, Time Series Analysis, A/B Testing

Other competencies: Predictive Analytics, Time Series Forecasting, PowerBI, Optimization, CI/CD, Git

PROJECTS

AI-Powered End-to-End Used Car Pricing Platform [\[GitHub Link\]](#)

Jul 2024 – Oct 2024

- Built an AI-driven platform to predict used car prices, integrating Flask, PostgreSQL, and ML for accurate valuation.
- Implemented web scraping with BeautifulSoup and Selenium to collect real-time vehicle listings, improving predictions.
- Developed an efficient data pipeline for preprocessing and storage, enhancing accuracy by 15% and improving recommendation relevance.

Fine-Tune FLAN-T5 for Less-Toxic Summarization using RL and PEFT [\[GitHub Link\]](#)

Jan 2025 – Mar 2025

- Developed and **fine-tuned** a FLAN-T5 model using **Proximal Policy Optimization (PPO)** and Parameter-Efficient Fine-Tuning (PEFT) on **AWS SageMaker** to reduce toxicity in generated summaries, resulting in a 25% improvement in toxicity scores.

CERTIFICATIONS AND COURSES

Certified Analytics Associate, Tiger Analytics [\(Link\)](#)

Machine Learning, Stanford University (Coursera) [\(Link\)](#)

Deep Learning Specialization, Deeplearning.ai [\(Link\)](#)

Applied Data Science Specialization, IBM [\(Link\)](#)